Dynamics of periodic red blood cell suspensions in confined Poiseuille flows

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Dynamics and rheology of red blood cell (RBC) suspensions in confined Poiseuille flows are studied numerically using simplified vesicle and capsule models. By using periodic boundary conditions along flow direction, the RBC and its images are forced to align to the same lateral position, showing a single file with equal interdistance. Lateral migration of RBC is analyzed by increasing RBC concentration ϕ (defined as the area or volume fraction occupied by RBCs). A scaling law for the migration velocity as a function of RBC concentration is proposed. The intrinsic viscosity $[\eta]$ (the contribution of each RBC on effective viscosity) shows a nontrivial property with the increase of RBC concentration ϕ, due to the spatial organization of RBCs. We find that the overall behavior of the periodic suspension shares the same features as those obtained for a suspension for which the periodicity along the flow line is relaxed. For the periodic suspension the computational time is greatly reduced as compared to a situation without imposing periodicity. This study offers an interesting perspective for the study of concentrated suspension, not only for a single file but also for more complex structures, where in each periodicity more than a single RBC is considered. These findings can help understanding the behaviors of blood flow in microcirculation with a highly reduced computational cost.

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I. INTRODUCTION

Red blood cells (RBCs) occupy about 45% of blood volume, and therefore determine the overall blood flow properties. In microcirculation (arterioles, capillaries, and venules), blood does not flow like Newtonian fluids but shows peculiar behaviors due to the presence of RBCs. For example, hematocrit is reduced in smaller and smaller vessels, known as the Fahraeus effect [1–3]. Furthermore, the apparent viscosity of blood decreases with decreasing vascular diameter, a phenomenon known as the Fahraeus-Lindqvist effect [3,4]. It is also known that in micro vessels, RBCs tend to form reversible clusters (which form and dissociate in the course of time) [5,6] under physiological conditions (e.g., for a normal concentration of fibrinogen, a plasma protein). In contrast, under pathological situations, such as in diabetes [7,8], these RBC clusters can become irreversible, impairing blood perfusion, and affecting oxygen delivery to tissues and organs [9,10] and biochemical signaling in microcirculation [11,12].

The origin of the Fahraeus-Lindqvist effect [3,4] is explained by the presence of a cell free layer (CFL) adjacent to the vessel wall, caused by a lateral migration of RBCs as they flow in vasculature. Using a simplified vesicle model (fluid enclosed by an inextensible phospholipid membrane), the lateral migration has been widely studied. For a single vesicle, there are two main situations that lead to lateral motion: (i) the wall effect which repels the vesicle away from the wall [13,14] and (ii) the shear gradient (or flow curvature) [15–17]. Note that the latter turned out to be more subtle [18,19]:

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it can either push vesicles toward the center of the flow or away from it, depending on the flow and vesicle properties. Moreover, the behaviors mentioned above have been observed numerically when using capsule models [20,21], indicating that lateral migration is an intrinsic characteristic of soft particles such as RBCs, regardless of their membrane properties.

The subtlety of lift due to flow curvature may lead to asymmetric slipper shapes [19,22,23]. In addition, simulations revealed a richer variety of off-centered RBCs in the presence of solid boundaries (i.e., confined Poiseuille flow), including stable slipper shapes [24], periodically oscillating shapes (snaking and swirling) [25], multilobes [26], and chaotic dynamics [27,28]. Asymmetric shapes have also been reported experimentally [29–32], indicating the complexity of blood flow in microcirculation.

The above studies have focused on a single entity. In real situations, RBCs have a finite concentration and therefore hydrodynamic interactions among cells may significantly contribute to cross-streamline migration. For dilute enough RBC suspensions, the hydrodynamic interaction is weak, and numerical simulations show that they form a single file around the center line [26,33]. By increasing the concentration, the hydrodynamic interaction among RBCs increases as the cell-cell distance is reduced. These hydrodynamics interactions may play a decisive role. For example, in confined shear flow, a breakdown of single file into ordered two-file solution, or disordered configuration has been reported [33,34]. A similar phenomenon is observed in confined Poiseuille flow as well [26], but has not been studied systematically yet.

How cross-streamline migration depends on interaction among cells is still not fully understood. For example, it needs to be clarified whether the CFL is enhanced or reduced due to interaction, and how it evolves with increasing hematocrit. To capture basic phenomena, we will make a simplification at the benefit of a significant reduction of computational time. We consider that the suspension is ordered in the flow direction and forms a single file (we consider confined enough suspensions), by adopting periodic boundary conditions. The elementary numerical box (containing a single cell) will be repeated periodically, and the suspension concentration will be monitored by varying the box length L: decreasing L corresponds to increasing concentration.

We will see that increasing concentration leads to a symmetry-breaking bifurcation where RBCs become off-centered and analyze in detail this bifurcation. Besides the static configuration analysis, we also analyze the behavior of the migration velocity as a function of concentration. We provide an analytical scaling law for the lateral migration velocity which fits remarkably well with the numerical simulation. Finally, the impact of hydrodynamic interaction on rheology is studied, showing that the intrinsic viscosity can be simply related to the RBC thickness along the spanwise direction. We will see that the main results (especially the bifurcation toward a noncentered solution, and the behavior of the effective viscosity) are in agreement with the full simulation without imposing periodicity. This opens a new way to tackle concentrated suspensions with various degrees of confinement by imposing periodicity, reducing thus considerably the computational time.

II. METHODS

A. Mathematical model

In this work, two widely used models are considered as representatives of RBCs: a two-dimensional (2D) vesicle model which accounts for membrane bending elasticity and a three-dimensional (3D) capsule model which accounts for both bending elasticity and in-plane shear elasticity. We confine ourselves mainly to the 2D vesicle model for the sake of numerical efficiency, as we have already shown that the 2D vesicle model captures the essential features of the 3D RBC (such as shapes under flow [27], cross-streamline noninertial migration velocity [15,16], inclination angle of cell with respect to flow direction [35,36], multilobe shapes [37], and ATP release under flow [38]). The 2D vesicle model is used as a faster tool to explore cell dynamics in wider ranges of parameters, while the 3D capsule model is used as complement to show the same trends, further validating the results.

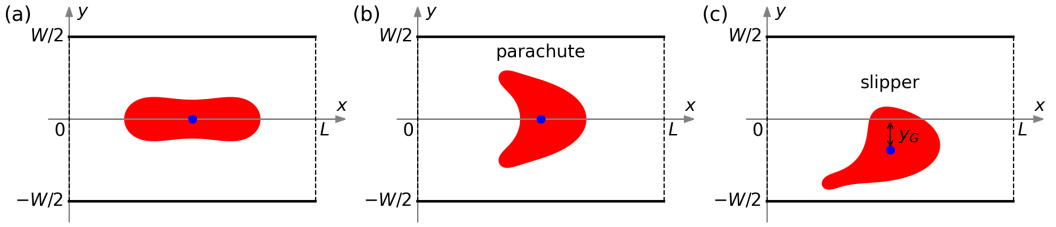


FIG. 1. Schematics of the studied configuration and steady-state shapes: (a) initial shape, (b) centered parachute, (c) off-centered slipper. L and W are the channel length and width, respectively. The blue dot denotes the mass center of the RBC. It should be noted that different initial positions are investigated, not just the configuration shown in panel (a). y_G in panel (c) denotes the lateral position of the RBC.

The schematic of the studied configuration is shown in Fig. 1(a). The channel length and width in 2D are L and W, respectively. In 3D, the channel is a cylindrical tube of length L and diameter W. The center line of the channel coincides with the x axis. Periodic boundary conditions are employed at the left and right boundaries $x = 0$ and $x = L$. A RBC is immersed in the straight channel, initialized as its equilibrium shape with its major axis parallel to flow direction. Its initial position can be different from that shown in Fig. 1(a). For a 2D RBC with perimeter P and area A, its characteristic radius is defined as $R_0 = \sqrt{A/\pi}$, which equals the radius of the circle with the same area. Its reduced area is defined as $\tau = 4\pi A/P^2$ to measure the degree of deflation. Similarly, for a 3D RBC with surface area S and volume V, its characteristic radius is defined as $R_0 = \sqrt[3]{3V/4\pi}$, which equals to the radius of the sphere with the same volume. Its reduced volume is defined as $\tau = 3V/4\pi/(S/4\pi)^{3/2}$. In the present work, we set $\tau = 0.65$ for all cases, inspired by that of RBC. An undisturbed Poiseuille flow $\boldsymbol{u}_0(u_{0x}, u_{0y})$ is imposed by a pressure gradient ∇p_0 as

$$u_{0x} = u_{\max}\left(1 - \frac{4r^2}{W^2}\right), u_{0y} = 0. \tag{1}$$

Here, $u_{\max}$ refers to the maximum flow velocity at the center line, and $r = y$ or $\sqrt{y^2 + z^2}$ in 2D or 3D, respectively.

In 2D, a RBC can be treated as a vesicle enclosed by a lipid membrane endowed with a bending force which follows the Helfrich bending energy [39]

$$E_{2D} = \frac{\kappa}{2}\oint c^2 ds + \oint \zeta ds, \tag{2}$$

where s is the curvilinear coordinate of the membrane contour, c is the membrane curvature, and ζ is the Lagrange multiplier that enforces local membrane inextensibility. The membrane force can be obtained by functional derivative of the above energy [15] as

$$\boldsymbol{F}_{2D}(s) = \kappa\left(\frac{\partial^2 c}{\partial s^2} + \frac{1}{2}c^3 - \zeta c\right)\boldsymbol{n} + \frac{\partial \zeta}{\partial s}\boldsymbol{t}, \tag{3}$$

where $\boldsymbol{n}$ and $\boldsymbol{t}$ are the normal and tangential vectors of the membrane.

The 3D capsule model consists of an in-plane deformation energy according to the Skalak law [40] together with the Helfrich bending energy

$$E_{3D} = \frac{G_s}{4}\oint (I_1^2 + 2I_1 - 2I_2 + CI_2^2)ds + \frac{\kappa}{2}\oint c_H^2 ds, \tag{4}$$

where G_s is the shear modulus, C is the energy penalty for area dilation, I_1 and I_2 are the strain invariants, and c_H is the mean curvature. Previous studies have suggested that the reference state of RBC membrane could be inhomogeneous [41,42], which leads to oscillating dynamics of RBC [43,44]. To avoid the effect of inhomogeneous reference state on lateral migration, in this

work, the reference state for the Skalak model is taken to be the sphere with the same surface area as the capsule. The membrane force are computed as described by Gou *et al.* [45].

The fluid dynamics is governed by the Navier-Stokes equation. The fluid velocity $\boldsymbol{u}$ and the pressure p are solved for by either the lattice Boltzmann method (LBM) in 2D or the finite volume method (FVM) in 3D. The immersed boundary method [46] is adopted to extract the interaction between RBC and fluid. In numerical application, the flow in the channel is imposed by applying a position-independent volume force everywhere in the fluid, which is equal to the pressure gradient ∇p_0. When using periodic boundary conditions, whatever flows out of the domain on the right boundary at $x = L$ re-enters the domain instantaneously at the left boundary at $x = 0$, so that

$$\boldsymbol{u}(\boldsymbol{x}, t) = \boldsymbol{u}(\boldsymbol{x} + \boldsymbol{L}, t),\ p(\boldsymbol{x}, t) = p(\boldsymbol{x} + \boldsymbol{L}, t), \tag{5}$$

where $\boldsymbol{L} = (L, 0, 0)$ is the vector of the periodicity direction. In the LBM, this is realized by introducing virtual lattice points before and after the periodic boundaries, and the distribution function are copied into these lattice points from the opposite periodic edge of the system before the streaming step [47]. In the FVM, the discretization process treats the periodic boundary faces as internal, while the neighbor cell of a near-boundary cell is the corresponding cell on the opposite periodic boundary [48]. These periodic boundary conditions are equivalent to an infinite channel in which the RBC, the fluid flow, and the pressure are all replicated along the flow direction. Hence, the flow in the computational domain is affected by the RBC and its images through hydrodynamic interactions. The detailed implementation of these numerical methods and their validations can be found in Shen *et al.* [34] and Gou *et al.* [49], respectively.

Note that thermal fluctuations are not included for either RBC membrane or fluid dynamics in this work. Indeed, thermal fluctuations are usually introduced for RBCs under low external load, resulting in cell diffusion [50] and membrane flickering [51]. At high external load such as adhesion and flow, the effect of thermal fluctuations is suppressed and is of minor importance [52,53]. In the study of RBC dynamics in shear flow, the simulations using smooth dissipative particle dynamics (SDPD) with thermal fluctuations and using IBM without thermal fluctuations show quantitative agreements [54]. Another study has also concluded that a 2D vesicle model without thermal fluctuations is able to capture the essential physical features of RBCs in flow [55]. The main purpose of this paper is to show that a periodic array of RBCs is able to capture essential features of multiple cells.

B. Dimensionless parameters

Now we introduce dimensionless parameters to describe the properties of the membrane and the flow field. The length and width of the channel are normalized by the characteristic radius R_0. The migration velocity is normalized by the maximum imposed flow velocity $u_{\max}$. The capillary number characterizes the ratio between the flow strength and membrane bending as

$$\mathrm{Ca} = \frac{\mu_{\mathrm{ex}} \dot{\gamma}_w R_0^3}{\kappa}, \tag{6}$$

where μ_{ex} is the viscosity of the suspending fluid, and $\dot{\gamma}_w = 4u_{\max}/W$ is the shear rate of the imposed flow at the channel wall. One could also define another capillary number for a 3D RBC related to the shear elasticity as

$$\mathrm{Ca}_s = \frac{\mu_{\mathrm{ex}} \dot{\gamma}_w R_0}{G_s}. \tag{7}$$

By keeping the dimensionless bending stiffness $\hat{\kappa} = \kappa/(G_s R_0^2)$ as a constant 0.03, Ca_s scales with Ca and will not be mentioned in the rest of the paper. In the present work, the capillary number is fixed at $\mathrm{Ca} = 90$ for most simulations, which guarantees symmetric shapes for RBCs in a long enough channel [27]. The RBC concentration is defined as the area fraction (in 2D) or the volume

fraction (in 3D) as

$$\phi = \frac{\pi R_0^2}{LW} \text{ (in 2D)}, \phi = \frac{16R_0^3}{3LW^2} \text{ (in 3D)}. \tag{8}$$

The ratio of the viscosity of the fluid inside the RBC over that of the suspending fluid is

$$\lambda = \frac{\mu_{\text{in}}}{\mu_{\text{ex}}}, \tag{9}$$

The effective viscosity of cell suspension is defined by

$$\mu_{\text{eff}} = \frac{\mu_{\text{ex}} Q_0}{Q}, \tag{10}$$

where Q is the average of the total flux (RBC + fluid), and Q_0 is the flux of the imposed flow given by Eq. (1). To evaluate the contribution of RBCs to the viscosity of the suspension, we define the intrinsic viscosity as

$$[\mu] = \frac{\mu_{\text{eff}} - \mu_{\text{ex}}}{\mu_{\text{ex}}\phi}. \tag{11}$$

III. RESULTS AND DISCUSSION

In this section, RBC dynamics and rheology in channels with different lengths are studied numerically. Since periodic boundary conditions are used in the flow direction, the single RBC and its images can be considered as a train of RBCs in an infinite channel. The effects of RBC concentration, channel confinement and viscosity contrast on the migration and final state of RBCs are explored. Diverse steady states and phase diagrams due to the hydrodynamic interaction among RBCs will be presented.

A. Symmetry breaking due to hydrodynamic interaction

We first focus on RBC dynamics in a highly confined channel ($W = 2.5R_0$). The viscosity contrast is set as $\lambda = 1$, and the effects of a different value will be discussed later. The initial position of RBC lies in the interval $[0, 0.625R_0]$ from the center line. By varying the channel length ($L \in [3.5R_0, 15R_0]$), the inter-cell distance and hydrodynamic interaction change correspondingly. In the current set-up, the solid boundaries tend to impose a centered parachute, as shown in Fig. 1(b). We will see that a strong enough mutual interaction between RBCs will lead to an instability of the centered parachute solution in favor of an off-centered sliper solution [see Fig. 1(c)].

1. Bifurcation of symmetric steady state (parachute)

We find a transition from a centered position, $y_G = 0$, to an off-centered position, $y_G \neq 0$, around $L \approx 8.2R_0$ ($\phi \approx 0.153$) at Ca = 90, as shown in Fig. 2(a). Here, y_G is the mass center of the RBC. In a long channel ($L \geqslant 8.2R_0$), the hydrodynamic interaction is weak, and the RBC shows a symmetric parachute shape ($y_G = 0$). With the decrease of channel length, the hydrodynamic interaction becomes stronger and eventually leads to the off-centered solution. The break down of symmetry indicates a supercritical bifurcation as a function of hydrodynamic interaction among RBCs, as shown in Fig. 2(b). We also performed simulations using a fixed channel length, $L = 30R_0$, within which several RBCs are inserted (without imposing periodicity within the length $L = 30R_0$; the suspension is periodic only with periodicity L). We then study the suspension dynamics by varying the the number of RBCs within box L. Within box L no periodic condition is imposed on RBCs; they are free to evolve without respecting periodicity. Still, the obtained positions for each RBC show the same bifurcating behavior (see the blue squares in Fig. 2). This clearly highlights that imposing periodicity does not impact the overall behavior.

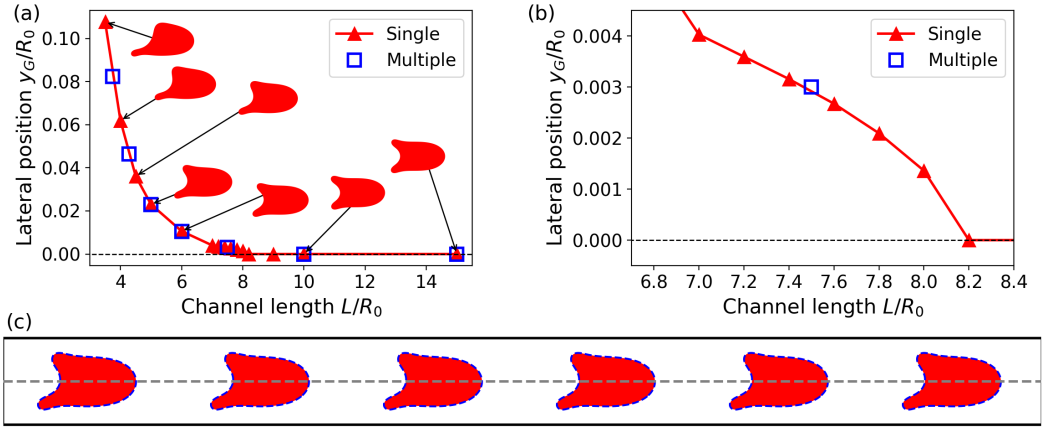


FIG. 2. (a) Steady-state lateral position of RBC y_G (in units of R_0) as a function of the channel length L (in units of R_0) at $\lambda = 1$ and $W = 2.5R_0$. Insets show steady-state RBC shapes at certain channel length. For the cases using multiple RBCs (blue squares), the relative channel length is L/nR_0, where n is the number of RBCs. (b) Zoom around the bifurcation point. (c) Snapshot of the RBC and its images at $L = 5R_0$ ($\phi = 0.251$). The blue dashed lines are the RBC configuration for the case using six RBCs in a box $L = 30R_0$ ($\phi = 0.251$).

To further confirm the supercritical bifurcation, we consider a quasistatic approximation, in that the mass center of the RBC is fixed until its shape is relaxed to a steady state for this fixed position, and then calculate the migration velocity $\dot{y}_G$ (see Farutin and Misbah [19] and the Supplemental Material [56]). By this method, all possible steady-state positions, including stable and unstable ones, can be captured (unlike the case where the center of mass is free to move, in which case only the stable solution can be captured). It should be noted that the sign of the migration velocity $\dot{y}_G$ represents its direction, that is, the velocity toward the center line is defined as negative. As shown in Fig. 3, the result obtained using quasistatic approximation matches well with those obtained by measuring instantaneous migration velocity of a freely moving RBC, except for the initial phase (i.e., $y_G > 0.5R_0$). For a freely moving RBC, if its shape deformation is much faster than its lateral

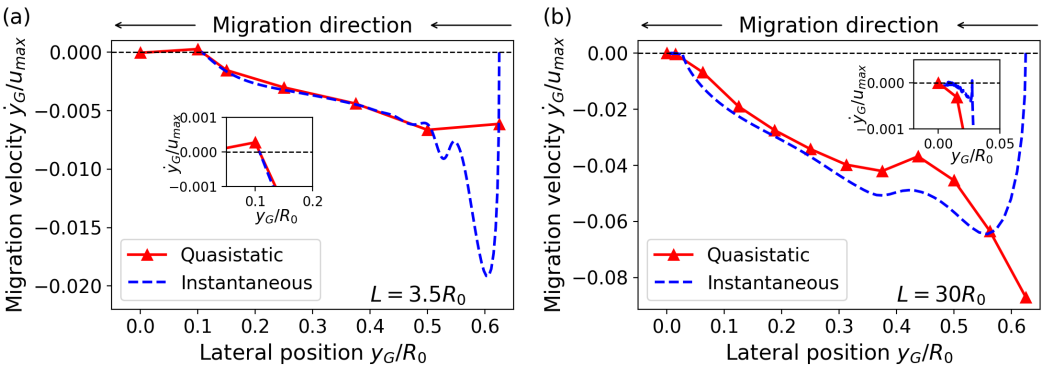


FIG. 3. Migration velocity $\dot{y}_G$ (in units of $u_{\max}$) obtained by quasistatic approximation (red triangle) and instantaneous simulation of a freely moving RBC (blue dash line) as a function of the lateral position y_G (in units of R_0). Insets show the curves in the vicinity of the stable steady state. The arrow indicates the migration direction of the freely moving RBC. The initial position of the RBC is $0.625R_0$ from the center line. The channel length is (a) $L = 3.5R_0$ ($\phi = 0.359$) and (b) $L = 30R_0$ ($\phi = 0.0419$), respectively. The channel width is $W = 2.5R_0$, and the viscosity ratio is $\lambda = 1$.

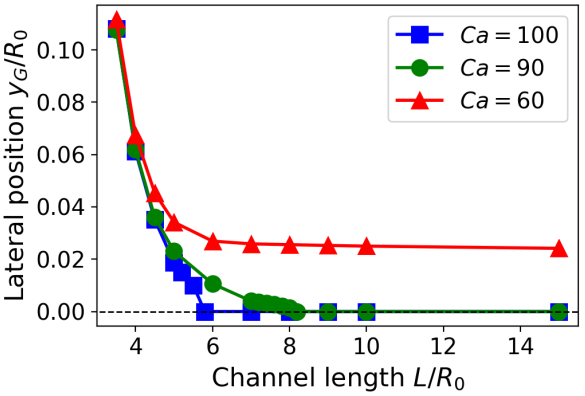


FIG. 4. Effect of capillary number Ca on steady-state lateral position y_G (in units of R_0) at $W = 2.5R_0$ and $\lambda = 1$.

motion, the instantaneous migration velocity would be close to that of the quasistatic migration velocity. During the initial phase ($y_G > 0.5R_0$), the RBC starts to deform from a biconcave shape [see Fig. 5(a)], which is different from the steady-state slipper shape obtained using quasistatic approximation [see Fig. 11(b)]. The instantaneous migration velocity is assumed to be 0 at the beginning, and its shape deformation rate is comparable to its lateral migration rate before it eventually deforms to a slipper. For this reason, there is a mismatch for the migration velocity at large y_G. When $L = 3.5R_0$ [$\phi = 0.359$, Fig. 3(a)], the quasistatic approximation indicates two steady-state positions: one off-centered steady-state solution located at the final position of the freely moving RBC, and another centered steady state located at $y_G = 0$. The stability of steady state depends on $d\dot{y}_G/dy_G$: If $d\dot{y}_G/dy_G$ is positive, then the steady state is unstable; otherwise, it is stable. Hence, the centered steady state is unstable, while the other one (the off-centered steady state) is stable. Due to the symmetry of geometry, the off-centered steady state appears in pairs, on both sides of the center line. For a long channel [$L = 30R_0$, $\phi = 0.0419$, Fig. 3(b)], there exists only a single stable steady state at the center line. The transition from one centered steady state to two off-centered steady states corresponds to a supercritical pitchfork bifurcation.

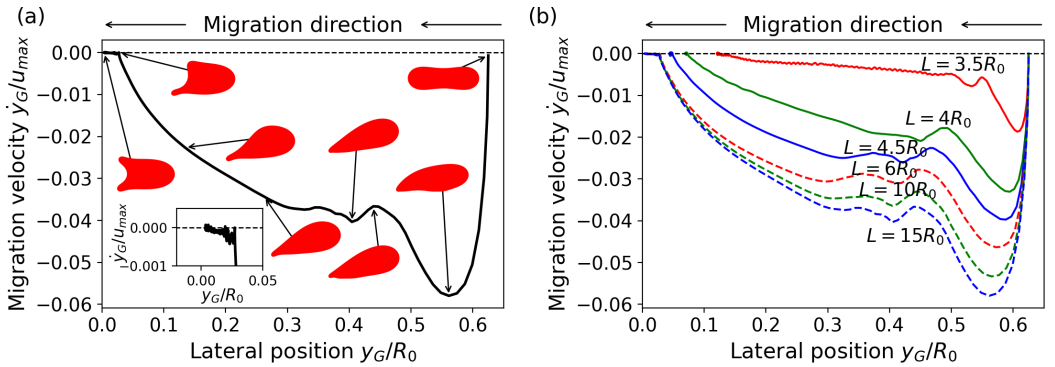


FIG. 5. (a) Instantaneous migration velocity $\dot{y}_G$ (in units of $u_{\max}$) as the function of the lateral position y_G (in units of R_0). The migration direction is denoted by arrows. Insets show instantaneous RBC shapes at chosen lateral positions. The channel length is $L = 15R_0$ ($\phi = 0.0838$). (b) Instantaneous migration velocity as the function of the lateral position at different channel length. The channel width is $W = 2.5R_0$, and the viscosity ratio is $\lambda = 1$.

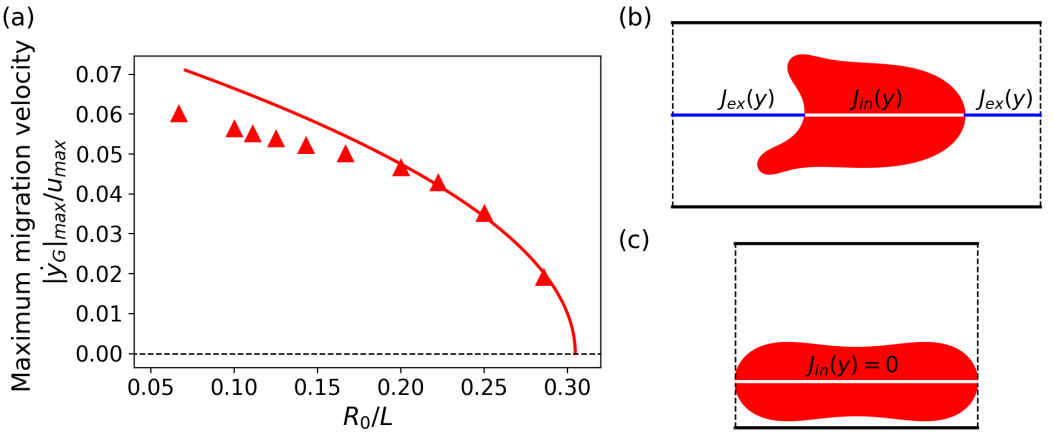


FIG. 6. (a) Maximum migration velocity in the first phase (see Fig. 5) $|\dot{y}_G|_{\max}$ (in units of $u_{\max}$) as a function of the reciprocal of channel length $1/L$ (in units of $1/R_0$). The solid lines are fitted curves with $(1/L_c - 1/L)^{1/2}$. (b, c) Schematics of flux in y direction. The white (blue) lines indicate the points inside (outside) RBCs.

Previous studies have shown the transition from parachute to slipper with decreasing capillary number, for a single RBC in either unconfined or confined Poiseuille flow [22,27]. Similar results are obtained in Fig. 4, and we further analyzed how the steady state evolves at different Ca. As shown in Fig. 4, the bifurcation of symmetric steady state is observed at Ca = 100 as well. However, the off-centered slipper emerges in shorter channel when compared to that at Ca = 90. When using lower capillary number (Ca = 60), the RBC remains off-centered even in a long channel ($L = 15R_0$). Hence, no symmetry-breaking presents, and the steady-state lateral position migrates toward the channel wall as the channel length is decreased. As the purpose of this work is to study symmetry breaking of RBCs, the capillary number is fixed at Ca = 90 if it is not specifically mentioned hereafter.

2. Decrease of migration velocity with increasing RBC concentration

The snapshots of shape evolution and migration velocity of a freely moving RBC at $L = 15R_0$ ($\phi = 0.0838$) are shown in Fig. 5(a). As stated above, the sign of the migration velocity represents its direction, where the velocity toward the center line is defined as negative. Hereafter an increase or decrease of velocity refers only to its magnitude (absolute value). As shown in Fig. 5(a), the lateral migration of RBC can be divided into three phases: (i) in the first phase ($y_G > 0.45R_0$), the RBC deforms amply, while its migration velocity first increases (in magnitude) and then decreases; (ii) in the second phase ($0.03R_0 < y_G < 0.45R_0$), the RBC gradually evolves into a slipper shape, with a decrease in the migration velocity magnitude; (iii) in the third phase ($y_G < 0.03R_0$), the RBC migrates toward the center line with a nearly zero velocity [see the zoom in Fig. 5(a)], and eventually assumes a parachute shape. By decreasing the channel length, the migration velocity decreases as well, and the maximum velocity magnitude is achieved at a position closer to the channel wall, as shown in Fig. 5(b). Note that the third phase is absent when the steady state is asymmetric, since the RBC stops at the fixed point corresponding to the end of phase (ii).

In this section, the channel width is set to a constant. To further dig into the effect of RBC concentration, we plot the maximum migration velocity in the first phase $|\dot{y}_G|_{\max}$ as a function of the inverse of the channel length $1/L$ ($\phi \sim 1/L$, normalized by $1/R_0$), as shown in Fig. 6(a). A curve fitting leads to, with a reasonable precision, to a scaling $|\dot{y}_G|_{\max} \sim (1/L_c - 1/L)^{1/2}$, where L_c is the critical channel length corresponding to zero migration velocity [see the solid line in Fig. 6(a)]. For the present channel width ($W = 2.5R_0$), the critical channel length is $L_c \approx 3.3R_0$ (corresponding critical RBC concentration $\phi \approx 0.381$), which is approximately equal to the RBC length in the

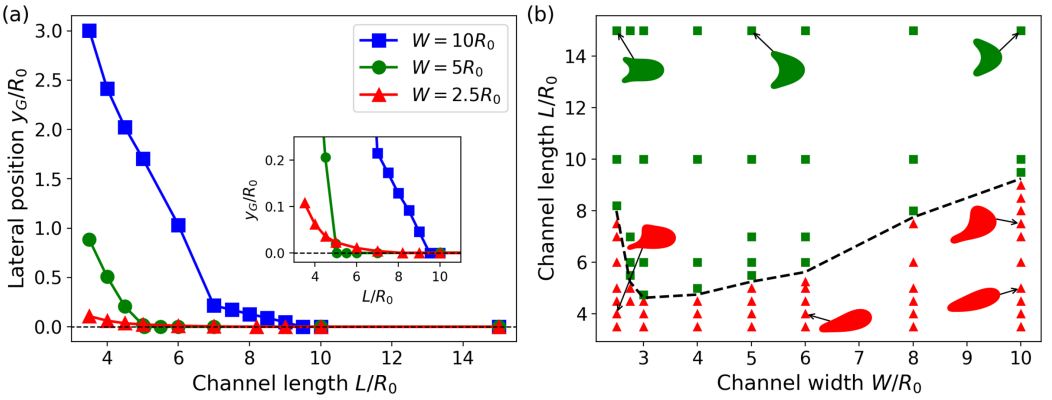


FIG. 7. Effects of channel width W on steady state: (a) steady-state lateral position of RBC y_G (in units of R_0) as a function of the channel length L (in units of R_0). The inset shows a zoom around bifurcation points. (b) Phase diagram of the steady state of RBC: off-centered slipper (red triangle) and on-centered parachute (green square). The dashed line denotes the bifurcation point L_b (in units of R_0) as a function of the channel width W (in units of R_0).

equilibrium state. The decrease of migration velocity is explained as follow. We first define the flux in y direction as

$$J(y) = J_{\text{in}}(y) + J_{\text{ex}}(y) = \int_0^L u_y(x, y) \cdot H(x, y)dx + \int_0^L u_y(x, y) \cdot (1 - H(x, y))dx. \tag{12}$$

Here J_{in} and J_{ex} are the flux of the fluid inside and outside the RBC, respectively, and $H(x, y)$ is the Heaviside function [which is 1 if point (x, y) is inside the RBC and is 0 if it is outside; see Fig. 6(b)]. With this definition, the migration velocity can be written as the area average of the flux $J_{\text{in}}(y)$ inside RBC

$$\dot{y}_G = \frac{\int_0^W J_{\text{in}}(y)dy}{\pi R_0^2}. \tag{13}$$

For an incompressible flow, $J(y) = 0$ and $J_{\text{in}}(y) = -J_{\text{ex}}(y)$ at any y position. Let us consider the extreme case where the fluid domain in x direction is fully occupied by the RBC as shown in Fig. 6(c). Under this condition, we have $J(y) = J_{\text{in}}(y) = 0$, hence the migration velocity is 0. With the decrease of RBC concentration ϕ, there is flow between the RBC and its images, increasing the magnitude of $J_{\text{ex}}(y)$ and $J_{\text{in}}(y)$. Thus, the migration velocity increases as well. This means that when the cell-cell interaction becomes maximum (distance between RBCs tends to zero) the migration velocity vanishes.

B. Effects of channel width on hydrodynamic interaction

In the previous section, we have shown that increasing RBC concentration (decreasing channel length) tends to push the RBC away from the center line for the chosen channel width $W = 2.5R_0$ (see Fig. 2). The question naturally arises of how does the bifurcation structure evolve with the channel width. A wide range of channel widths is investigated, namely $W \in [2.5R_0, 10R_0]$. All other parameters values are kept identical to those of the previous section.

1. Local and global interactions between RBCs

For different channel widths, the transition from a centered to an off-centered position still behaves like a supercritical bifurcation with decreasing channel length, as shown in Fig. 7(a). The results are summarized in a phase diagram in the plane of channel length and channel width

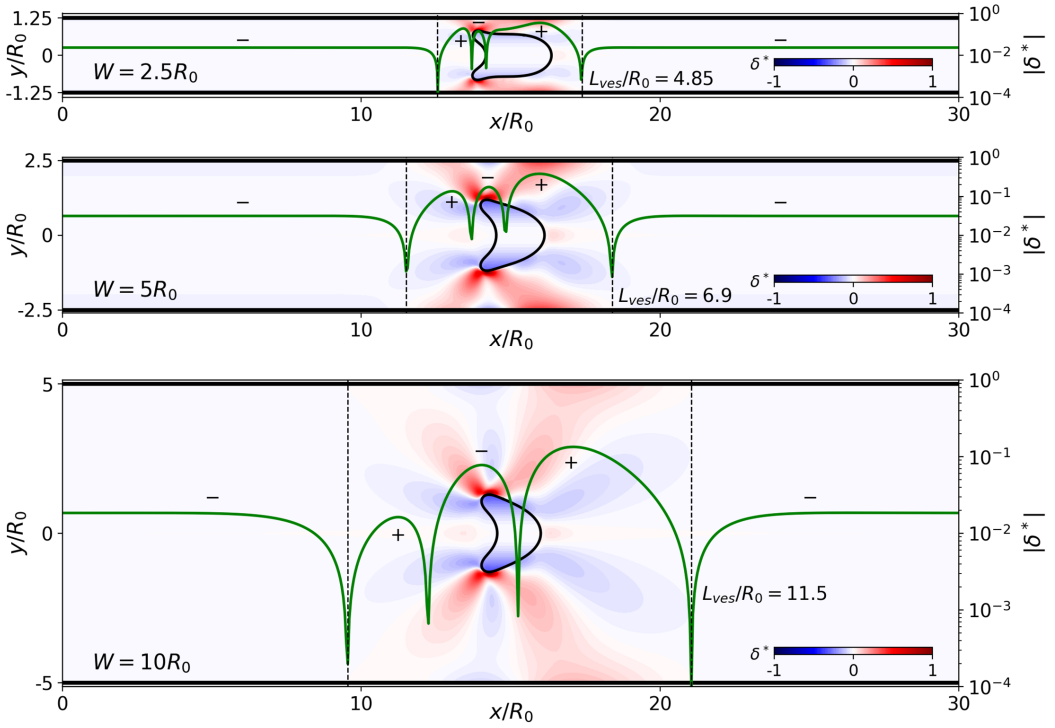


FIG. 8. Dissipation density in dimensionless unit δ^* around the RBC at $W = 2.5R_0$ (upper panel), $W = 5R_0$ (middle panel), and $W = 10R_0$ (lower panel). The solid black bold lines indicate the channel walls. The green solid curves are the absolute values of the dissipation density near the lower wall along flow direction, and their labels are on the right y axis. The plus and minus symbols above the green solid curves indicate the signs of δ^*. The dashed lines are the borders beyond which the RBC influence is negligible, and the local interaction range L_{ves} is the distance between the dashed lines.

[Fig. 7(b)]. The red symbols refer to the asymmetric solution whereas the green ones to the symmetric one. The bifurcation line L_b as a function of channel width W is also represented. Note that L_b reaches a minimum around $W \approx 3R_0$. This nonmonotonic behavior is due to an interplay between the local and global interactions between RBCs and their images, as explained below.

In Poseuille flow, the presence of RBC disturbs the flow field around it, which we refer to as local interaction between the RBC and its images. The dissipation density δ is used to evaluate the local interaction as

$$\delta = \tfrac{1}{2}\mu_{\text{ex}}|\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T|^2 - \tfrac{1}{2}\mu_{\text{ex}}\left|\nabla \boldsymbol{u}_0 + \nabla \boldsymbol{u}_0^T\right|^2. \tag{14}$$

Here the velocity $\boldsymbol{u}$ is composed of the imposed velocity $\boldsymbol{u}_0(u_{0x}, u_{0y})$ and the induced velocity due to the RBC $\boldsymbol{u}_{\text{ind}} = \boldsymbol{u} - \boldsymbol{u}_0$. For a better comparison between different cases, the dissipation density is normalized by its maximum value as $\delta^* = \delta/\delta_{\max}$. As shown in Fig. 8, the dissipation density reaches a maximum in the vicinity of the RBC. Far away, the dissipation density becomes almost constant, and its value is an order of magnitude smaller than the peak value. To quantify the local interaction range, we measure the dissipation density magnitude $|\delta^*|$ on the lower wall, as shown by the green solid lines in Fig. 8. Here the troughs result from the change of sign for δ^*, and the signs of δ^* are denoted by the plus and minus symbols. The distance between the two significant troughs before and after the RBC (denoted by the vertical dashed lines in Fig. 8) is defined as the local interaction range L_{ves}. With this definition, we find that the local interaction range L_{ves} increases linearly with the channel width as shown in Fig. 9(a). The screening of hydrodynamic

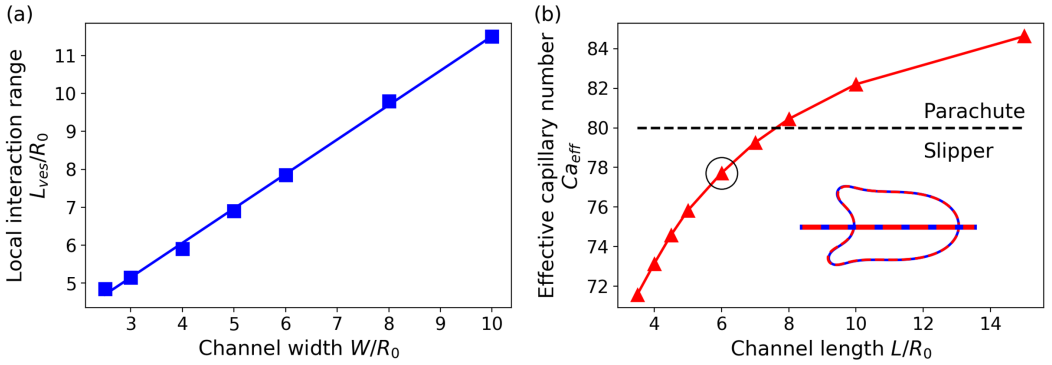


FIG. 9. (a) Local interaction range L_{ves} (in units of R_0) as a function of the channel width W (in units of R_0). The solid line is a fitted curve with linear function. (b) Effective capillary number Ca_{eff} as a function of the channel length L (in units of R_0) at $W = 2.5R_0$. The black dashed line denotes the critical capillary number for the transition from parachute to slipper. The inset shows snapshots and lateral positions of two RBCs at the same effective capillary number $\text{Ca}_{\text{eff}} \approx 78$ (labeled by the black circle): $\text{Ca} = 90$, $L = 6R_0$ for the red dashed line; $\text{Ca} = 80$, $L = 30R_0$ for the blue solid line.

interactions by channel confinement has been investigated by Thiebaud *et al.* [33]. This accounts for the behavior of the bifurcation point L_b when $W > 3R_0$. The increase of L_b in highly confined channel ($W < 3R_0$) is due to the increase of flow resistance, i.e., the global interaction. Indeed, with the decrease of channel length, the RBC concentration increases, leading to the increase of effective viscosity. As a consequence, the flow strength felt by the RBC and its images decreases. To quantify this value, we define the effective capillary number Ca_{eff} as

$$\text{Ca}_{\text{eff}} = \frac{6\mu_{\text{ex}}\bar{u}R_0^3}{W\kappa}, \tag{15}$$

where $\bar{u}$ is the mean velocity of RBC suspension. For a channel of infinite length, we have $\text{Ca}_{\text{eff}} = \text{Ca}$. In wide channels ($W \in [4R_0, 10R_0]$), the transition from parachute to slipper is found at a critical capillary number lower than 50 [27]. The change of Ca_{eff} with channel length is quite small, and the value of Ca_{eff} is always larger than the critical capillary number for shape transition. However, in highly confined channels ($W < 3R_0$), the critical capillary number increases with channel width [27]. For example, the critical capillary number is around 80 at $W = 2.5R_0$, as shown in Fig. 9(b). As the RBC diameter is comparable to the channel width, the value of Ca_{eff} decreases significantly as the channel gets shorter, and finally becomes smaller than the critical value for shape transition. For this reason, the steady state changes from parachute to slipper at the channel length $L \approx 8R_0$, larger than the local interaction range $L_{\text{ves}} \approx 5R_0$. We further show that at the same effective capillary number $\text{Ca}_{\text{eff}} \approx 78$, the RBC shows for both cases the same steady-state shape [see the inset of Fig. 9(b)]. This confirms the role of global interaction which is associated with an increase of the flow resistance.

2. Decrease of migration velocity with increasing channel width

At different channel widths, the migration velocity of a freely moving RBC can be divided into two or three phases, similar to those discussed in Fig. 5, while its magnitude becomes smaller. Recall that the inward migration (toward the center line) is induced by both the wall and shear gradient. Inspired by the migration laws in Farutin and Misbah [18], we can write the migration velocity of a RBC in an infinitely long channel in the following form:

$$\frac{\dot{y}_G}{u_{\max}} = \frac{C_1 R_0 y_G}{W^2} + \frac{C_2 R_0 y_G^3}{W^4}, \tag{16}$$

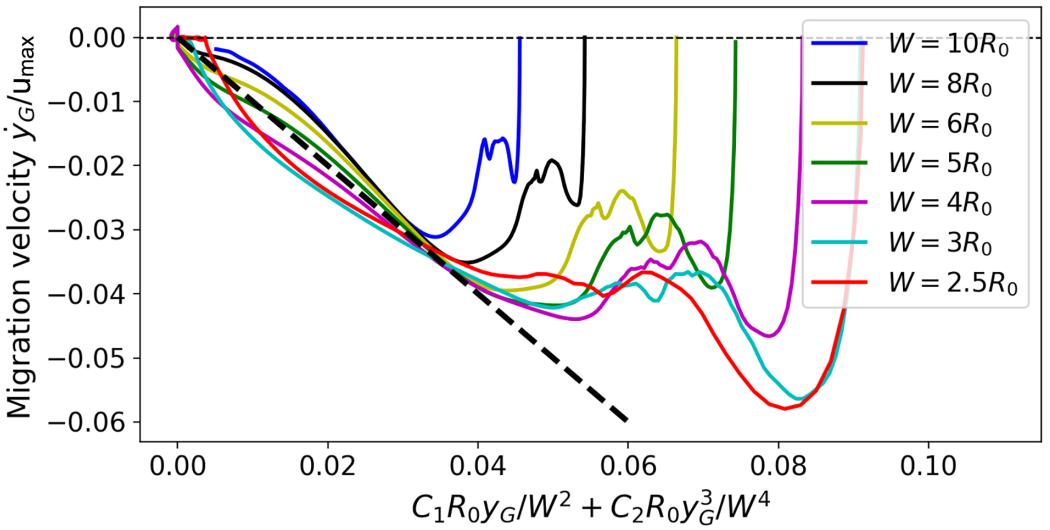


FIG. 10. Instantaneous migration velocity $\dot{y}_G$ (in units of u_{max}) for different channel widths W in a long channel ($L = 15R_0$). The black dashed line corresponds to Eq. (16).

where C_1 and C_2 are two negative constants determined numerically. As shown in Eq. (16), both terms on the right-hand side decrease as the channel width increases. Figure 10 shows the comparison between the migration velocity predicted by Eq. (16) and the instantaneous migration velocity obtained by numerical simulations. Here the channel length is $L = 15R_0$. Good agreements are found during the second phase of migration (see Sec. III A 2). The two constants are fitted as $C_1 = -0.85$ and $C_2 = -1.0$, respectively. The deviation might result from the variation of RBC shape with channel width.

The relation between migration velocity and the reciprocal of channel length is plotted in Fig. 11(a). The results are fitted with $|\dot{y}_G|_{max} \sim (1/L_c - 1/L)^{1/2}$. As the channel width increases, the critical channel length of zero migration velocity L_c increases as shown in Fig. 11(b). The increases

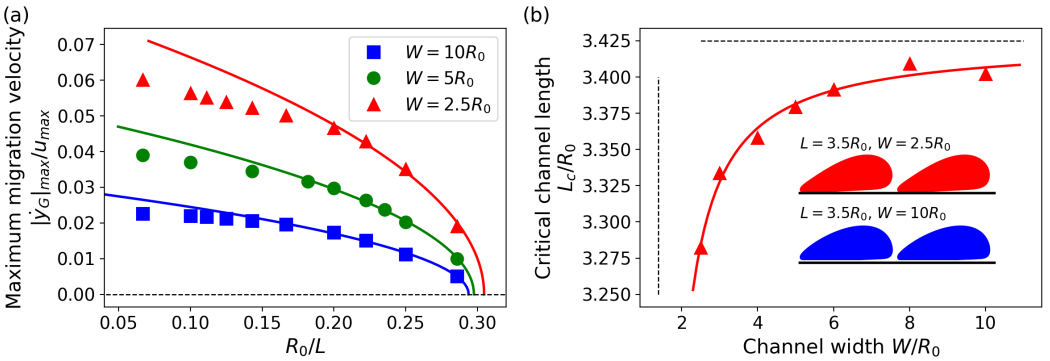


FIG. 11. Effects of channel width W on migration velocity: (a) Maximum migration velocity in the first phase $|\dot{y}_G|_{max}$ (in units of u_{max}) as a function of the reciprocal of channel length $1/L$ (in units of $1/R_0$). The solid lines are fitted curves with $(1/L_c - 1/L)^{1/2}$. (b) The critical channel length of zero migration velocity L_c (in units of R_0) as a function of the channel width W (in units of R_0). The solid line is a guide for the eyes, and the dashed lines are potential bounds. The insets show snapshots of RBCs by quasistatic approximation, of which the mass center is $0.625R_0$ away from the solid boundary.

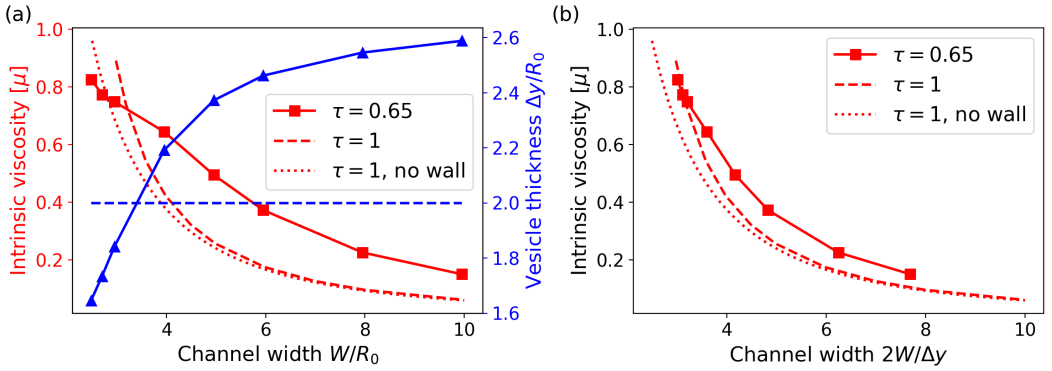


FIG. 12. Effects of RBC thickness on intrinsic viscosity in a long enough channel ($L = 30R_0$): (a) Intrinsic viscosity $[\mu]$ (red) and RBC thickness Δy (blue, in units of R_0) as functions of channel width W (in units of R_0). (b) Intrinsic viscosity $[\mu]$ as a function of channel width W (in units of $\Delta y/2$). The solid lines with symbols indicate RBC ($\tau = 0.65$), the dashed lines indicate rigid circular particle ($\tau = 1$), and the dotted lines indicate rigid circular particle when ignoring the interactions with solid boundaries ($\tau = 1$, no wall).

of L_c with channel width is due to the inclination of RBC, as shown in the inset of Fig. 11(b). In a less confined channel, the inclination angle is smaller, leaving less space for the RBC and its images. For this reason, the zero migration velocity emerges at larger L_c. Furthermore, the value of L_c should have an upper bound, as it cannot be longer than the length of a deformed RBC [see the horizontal dashed line in Fig. 11(b)].

Moreover, we speculate that there should be a lower bound for channel width below which there is no lateral migration [see the vertical dashed line in Fig. 11(b)]. For a sufficiently confined channel the RBC is in a close contact to the wall and we expect no lateral migration. Several experimental and numerical studies have reported on the case of very confined channels where the RBC almost completely occupies the cross section [57–59], hence without possibility of lateral migration.

C. Impacts on rheology

In this section, we analyze the rheology of the suspension. The intrinsic viscosity $[\mu]$ is used to quantify the contribution of RBCs to the suspension viscosity [see Eq. (11)].

1. Effects of RBC thickness

We first focus on the situation with a long enough channel (a very dilute regime), in which case the RBC is located at the center line and shows a parachute shape, as discussed before. The corresponding intrinsic viscosity as a function of channel width is plotted in Fig. 12(a) as the red squares. It is interesting to compare this result to numerical and analytical results of a rigid particle. For a rigid circular particle ($\tau = 1$) with a radius of R_0, when it is located at the center line, the intrinsic viscosity is shown by the red dashed line in Fig. 12(a). In the absence of bounding walls, the intrinsic viscosity is obtained analytically [$6R_0^2/W^2$, see the Appendix), as shown by the red dotted line in Fig. 12(a)]. One can conclude that the effect of channel boundaries is negligible for wide channels ($W > 6R_0$). While in narrow channels ($W < 4R_0$), the presence of channel boundaries leads to higher intrinsic viscosity. Note that there is no flow inside a centered parachute, hence the RBC flows like a rigid particle. The deviation between the numerical and analytical results is due to different deformation amplitude of the RBC for different channel widths. We define the RBC thickness Δy as the maximum cross section of the RBC in y direction, and plot its value as the blue triangles in Fig. 12(a). For the circular particle, its thickness is equal to its diameter $2R_0$ [see the blue dashed line in Fig. 12(a)]. In wide channels ($W > 3R_0$), the RBC has larger thickness than the circular particle, hence the hydrodynamic friction and intrinsic viscosity are also higher.

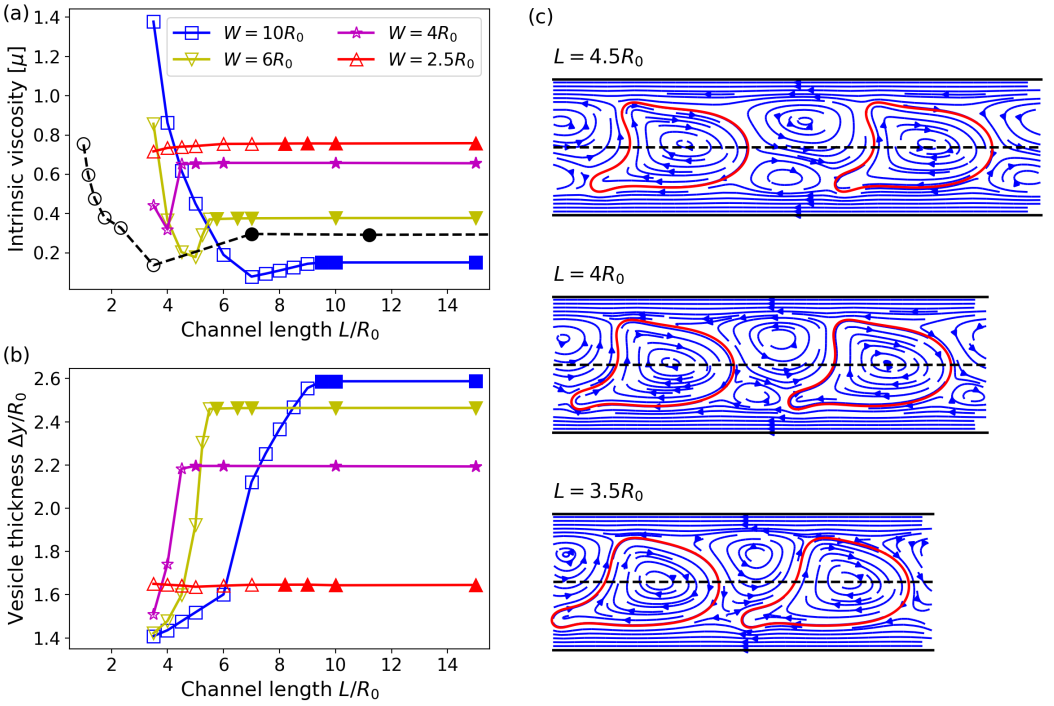


FIG. 13. Effects of channel geometry on (a) intrinsic viscosity $[\mu]$ and (b) vesicle thickness Δy. Empty symbols represent off-centered slipper shapes, while filled symbols represent on-centered parachute shapes. The black symbols are obtained from Fig. 10(a) in Ref. [26] when using multiple RBCs in a long channel ($L = 56R_0$, $W = 7R_0$, Ca $= 80$), and the relative channel length is L/nR_0, where n is the number of RBCs. (c) Steady-state RBC configurations and streamlines in a frame comoving with the RBC for $W = 2.5R_0$. The dashed lines indicate the center line of channel. In each configuration, the RBC and its image are plotted.

Conversely, in narrow channels ($W \leqslant 3R_0$) the situation is reversed: the RBC is compressed so that its thickness is smaller than the circular particle, resulting in lower friction and thus lower intrinsic viscosity. After rescaling the channel width W by the RBC thickness Δy, we find that the results for a deformable RBC are reasonably closer to those of the rigid circular particle (with or without channel boundaries), as shown in Fig. 12(b). Moreover, in highly confined channels ($W \leqslant 3R_0$), the intrinsic viscosity for RBC and rigid circular particle collapses. This suggests that the intrinsic viscosity depends on the particle thickness, regardless of whether the particle is deformable or not.

2. Nonmonotonic behaviors of intrinsic viscosity with RBC concentration

Here we consider the evolution of the viscosity with RBC concentration. For that purpose, we vary the channel length L. The evolution of the intrinsic viscosity $[\mu]$ with the channel length L is shown in Fig. 13(a). When the RBC shows a parachute shape (filled symbols), $[\mu]$ reduces to a constant value [see the plateaus in Fig. 13(a)]. After the symmetry breaking of the parachute shape [filled symbols in Fig. 13(a)], $[\mu]$ decreases in a quasilinear manner with the decrease of channel length (i.e., increase of RBC concentration). In a less confined channel ($W \geqslant 4R_0$), decreasing further the channel length leads to a second transition, in that that $[\mu]$ exhibits an increase. This second transition appears at smaller and smaller L as W decreases. We further compare the present results with those using multiple RBCs (i.e., without imposing periodicity on the scale of a single RBC) in a long channel ($L = 56R_0$, $W = 7R_0$, Ca $= 80$) [26]. Here the relative channel length is defined as L/nR_0, where n is the number of RBCs. As shown by the black symbols in Fig. 13(a), two

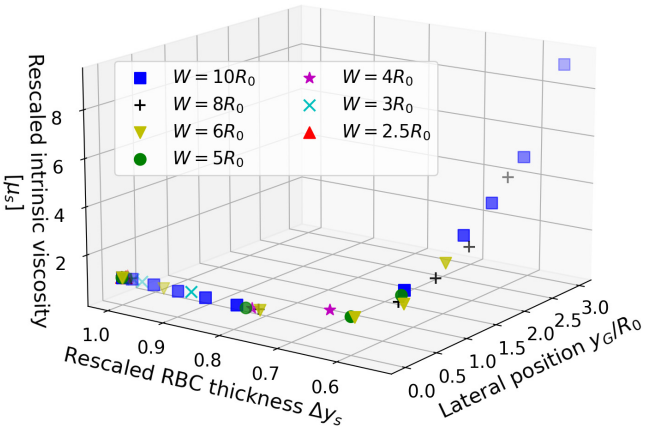


FIG. 14. Rescaled intrinsic viscosity $[\mu_s]$ as a function of the lateral position y_G (in units of R_0) and the rescaled RBC thickness Δy_s.

transitions of $[\mu]$ are observed, in qualitative agreement with the case where periodicity is imposed. It should be noted that the second transition occurs at higher RBC concentration (scales as nR_0/L) for the multiple RBCs case, because parachute and slipper shapes can appear simultaneously for multiple RBCs in a wide channel ($W = 7R_0$ in Ref. [26]).

The behavior of $[\mu]$ is affected by three different factors: (i) the lateral position, (ii) the RBC thickness, and (iii) the recirculation between the RBC and its images. First, the shear rate gets higher as the RBC migrates toward solid boundaries (empty symbols in Fig. 13, corresponding to slipper solution). This results in higher viscous dissipation, which leads to an increase of the intrinsic viscosity. Second, the thickness of a parachute RBC is larger than that of a slipper RBC [see Fig. 13(b)]. As the RBC migrates away from the center line, it occupies less cross section, reducing thereby the intrinsic viscosity. Finally, as the inter-cell distance decreases, the recirculation between the RBC and its images becomes weaker [see Fig. 13(c)], leading to a decline of the intrinsic viscosity [33,34]. The competition of these three factors results in the nonmonotonic behavior of $[\mu]$ with the channel length L. For a highly confined channel ($W \leqslant 3R_0$), either the lateral migration or the RBC lateral width (or thickness) is reduced [see Figs. 7(a) and 13(b)]. This leads to the fact that the intrinsic viscosity is dominated by the degree of recirculation and shows monotonic behavior [see the red line in Fig. 13(a)].

From the above analysis we can conclude that the contribution of recirculation is relatively small when compared to the other two factors. We thus analyze the behavior of the intrinsic viscosity $[\mu]$ with the lateral position y_G and the RBC thickness Δy. Moreover, as both $[\mu]$ and Δy have channel-width-dependent plateau values $[\mu]_{\text{inf}}$ and Δy_{inf} (see Fig. 12), the following scaling is suggested

$$[\mu_s] = \frac{[\mu]}{[\mu]_{\text{inf}}}, \Delta y_s = \frac{\Delta y}{\Delta y_{\text{inf}}}. \tag{17}$$

The rescaled intrinsic viscosity $[\mu_s]$ as a function of the lateral position y_G and the rescaled RBC thickness Δy_s reveals a reasonable data collapse, as depicted in Fig. 14. This finding indicates that the rheology of a RBC suspension is determined by RBC configurations, including their lateral positions and their extent in the spanwise direction. Indeed, the rescaled RBC thickness measures the lateral occupancy of RBC, which has been shown to be related to the viscosity. For a tumbling cell, its cross section oscillates periodically with its spanwise direction, so does the intrinsic viscosity [60]. A recent work further shows that the local fluidity is correlated with the local concentration, which is the space occupied by the cell over the channel width [61]. The resulted global viscosity could be obtained by integrating the local fluidity along the spanwise direction.

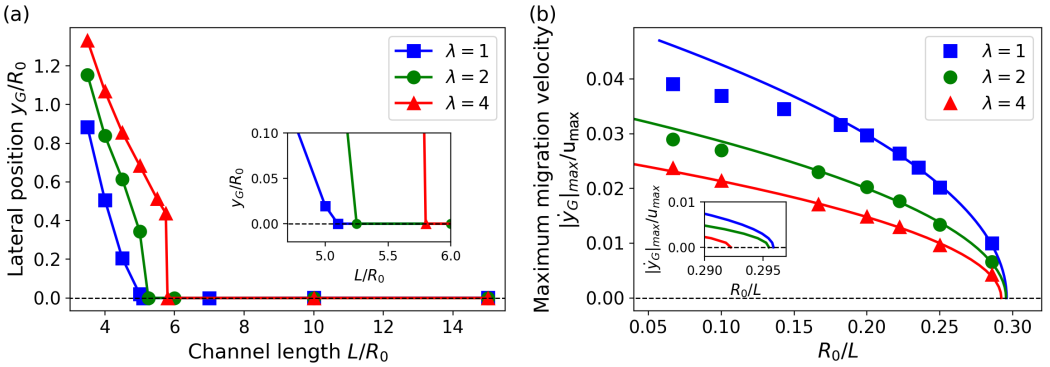


FIG. 15. Effects of viscosity contrast λ on lateral migration at $W = 5R_0$: (a) steady-state lateral position of RBC y_G (in units of R_0) as a function of the channel length L (in units of R_0). The inset shows a zoom around bifurcation points. (b) Maximum migration velocity in the first phase $|\dot{y}_G|_{\max}$ (in units of $u_{\max}$) as a function of the reciprocal of channel length $1/L$ (in units of $1/R_0$). The solid lines are fitted curves with $(1/L_c - 1/L)^{1/2}$. The inset shows a zoom around R_0/L_c.

D. Impact of viscosity contrast on rheology

Finally, we discuss the effects of viscosity contrast λ in this section. By increasing the value of λ, the migration toward solid boundary is enhanced [19,62]. As a consequence, the critical channel length for symmetry breaking is higher [see Fig. 15(a)]. However, the lateral migration velocity decreases [see Fig. 15(b)]. Moreover, the critical channel length of zero migration velocity L_c increases due to the decrease of inclination angle caused by a large viscosity contrast [63] [see Fig. 15(b)]. A systemic study on RBC dynamics at physiological viscosity contrast ($\lambda = 6$ [64]) is given in the Supplementary Material. More diverse steady states are exhibited and reported in a phase diagram (see Fig. S1 in the Supplemental Material [56]). A reduced lift at higher viscosity contrast is observed [65]. In highly confined channels ($W \leqslant 3R_0$), the off-centered slipper prevails (see Fig. S1 in the Supplemental Material [56]). This confined slipper is similar to that reported in Saadat *et al.* [66]. In less confined channels ($W \geqslant 4R_0$), we observe coexistence between the off-centered slipper and the parachute shape (see Figs. S1 and S2 in the Supplemental Material [56]), which have been also observed in other simulations and experimental studies [20,26]. Another shape, the multilobe one, emerges in wide channel ($W \geqslant 8R_0$) and coexists with the off-centered slipper (see Figs. S1 and S3 in the Supplemental Material [56]), in agreement with the numerical study of a single RBC in long channels [26].

The effect of viscosity contrast λ on intrinsic viscosity $[\mu]$ is shown in Fig. 16(a). When the RBC mass center is zero, $[\mu]$ is unaffected, since a parachute shape does not show tank-treading, meaning that the viscosity contrast is irrelevant. For an off-centered RBC, the intrinsic viscosity increases with viscosity contrast, due to tank-treading. Here again, the relation between lateral position y_G and reduced intrinsic viscosity $[\mu_s]$ is confirmed for different viscosity contrasts, as shown in Fig. 16(b).

E. Validation in 3D

To confirm that the observed phenomena are not a consequence of dimension but are due to an intrinsic property for RBCs, we further conduct simulations using a 3D capsule model. In this section, the RBC is initially put at a position very close to the center line of the channel. The capillary number is fixed at $\mathrm{Ca} = 45$, the channel length varies from $4R_0$ to $22R_0$, and the viscosity contrast varies from 1 to 3. The lateral position of RBC is defined as $r_G = \sqrt{y_G^2 + z_G^2}$, where y_G and z_G are the lateral positions in y and z directions. Two different channel widths are analyzed: $W = 3.55R_0$ and $7.09R_0$, which corresponds to a confinement of $Cn = 0.56$ and 0.28, respectively.

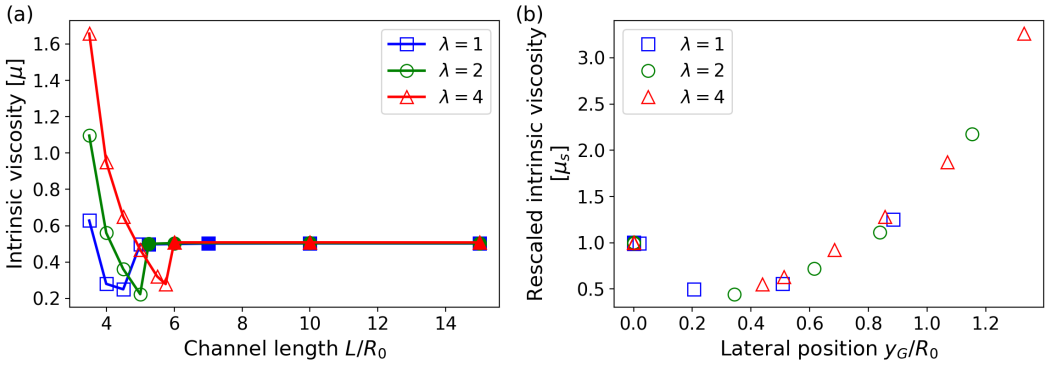


FIG. 16. Effects of viscosity contrast λ on rheology at $W = 5R_0$: (a) Intrinsic viscosity $[\mu]$ as a function of the channel length L (in units of R_0). Empty symbols represent off-centered slipper shapes, while filled symbols represent on-centered parachute shapes. (b) Rescaled intrinsic viscosity $[\mu_s]$ as a function of the lateral position y_G (in units of R_0).

For wide channels ($W = 7.09R_0$), symmetry breaking due to channel length is observed as shown in Fig. 17(a). Here the markers indicate time-averaged values, while the error bars indicate standard deviations. In long channels ($L > 15R_0$), the RBC presents a centered parachute shape, as shown in Fig. 17(b). By decreasing the channel length, the hydrodynamic interactions tend to push the RBC outward, leading to snaking motion around the center line of the channel ($L \in (12.5R_0, 15R_0)$, see Fig. S4 in the Supplemental Material [56]). Further decreasing the channel length results in off-centered slipper shapes ($L < 12R_0$), as shown in Fig. 17(c). It should be noted that we do not observe stable slipper shapes in 3D, unlike the cases using 2D vesicle model. Figure S4 in the Supplemental Material [56] shows the lateral positions and trajectories of snaking, swirling, and chaotic dynamics. As the spherical reference state is homogeneous with regard to cytoskeleton,

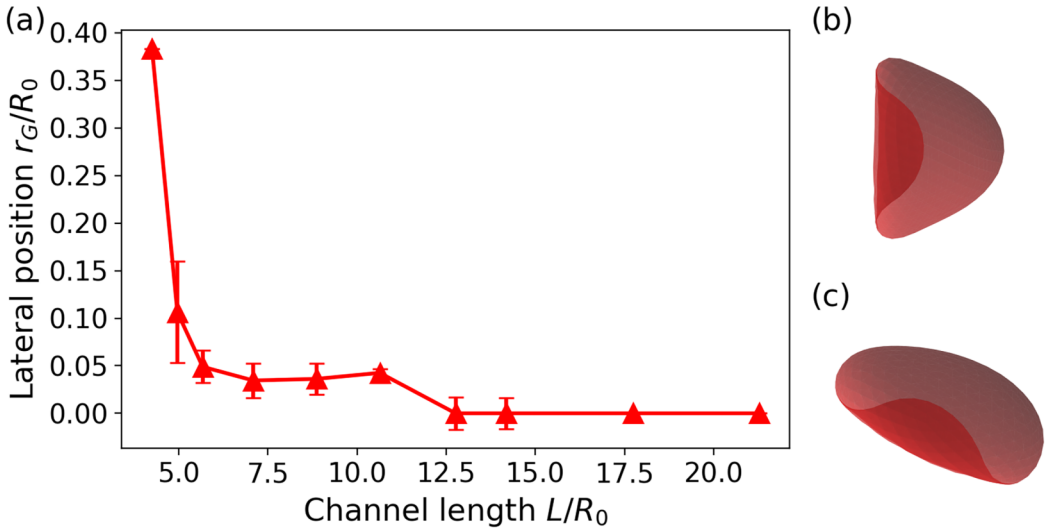


FIG. 17. Lateral migration of 3D RBCs in a cylindrical channel at $W = 7.09R_0$ and $\lambda = 2$: (a) Steady-state lateral position of RBC r_G (in units of R_0) as a function of the channel length L (in units of R_0). Here the markers indicate time-averaged values, and the error bars indicate standard deviations. Panels (b) and (c) show snapshots of RBC at $L = 21.28R_0$ and $L = 4.26R_0$, respectively.

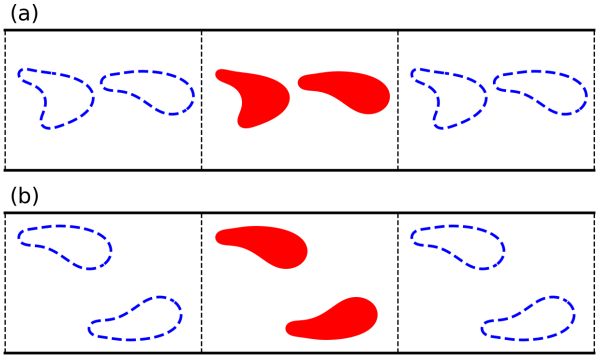


FIG. 18. Schematic of periodic RBC arrays in an elementary box: (a) unstable single file; (b) ordered two files. The blue dashed lines indicate the images of RBCs.

these oscillating dynamics are intrinsic properties of RBCs resulting from viscosity contrast. These oscillating dynamics are different from those reported by Sinha and Graham [44], which result from inhomogeneous reference states rather than viscosity contrast. For narrow channels ($W = 3.55R_0$), only parachute shapes are observed for different channel lengths and viscosity contrasts (not shown), indicating the strong wall-induced lift [24]. Despite the complex behaviors for 3D RBCs, the overall dynamics are in qualitative agreement with 2D results.

IV. CONCLUSIONS

We have investigated numerically the dynamical behavior and rheology of RBC suspensions in confined Poiseuille flows. By imposing periodic boundary conditions, we have focused on an idealized periodic suspension. This assumption leads to a significant reduction of computational time, since the simulation focused actually on a single box containing a single RBC. The effect of mutual interaction is studied systematically by varying the channel length. A symmetry breaking is observed with the decrease of channel length, indicating a supercritical bifurcation. The bifurcation point L_b shows a nonmonotonic behavior with increasing channel width, resulting from an interplay between the local and global interaction between the RBC and its images. The migration velocity is found to scale with RBC concentration a square-root function, and is enhanced by decreasing the channel width and the viscosity contrast. Moreover, we find data collapse when the viscosity is represented as a function of rescaled thickness, pinpointing the relevance of only this single dimensionless parameter as a relevant measure of rheology. The nonmonotonic behavior of intrinsic viscosity with RBC's lateral position and thickness indicates the existence of a universal law. However, a basic understanding of this behavior is at present missing and should require further studies. We have checked that relaxing the periodic nature of suspension (i.e., by solving for a set of a large number of RBCs in a box without imposing periodicity) captures the essential features exhibited by the forced periodic array. This opens a new way to handle the problem of rheology of concentrated suspensions with a much lower computational effort.

The purpose of this study is not to extract all possible RBC dynamics, but to show that periodic single array of RBCs with reduced channel length is able to capture essential features and similar behaviors as when the length is not reduced. The lack of shape variety is due to the limited parameter space chosen in this work. Despite this, we still observed oscillating dynamics at $\lambda > 1$, including multilobes, snaking, swirling, and chaotic dynamics (see Figs. S3 and S4 in the Supplemental Material [56]). In future works, we plan to explore more diverse RBC dynamics [28,44] with other reference states, capillary numbers, and confinements. The next step is to consider the elementary box filled by more than a single RBC with more complicated patterns, and imposing periodic arrays to tackle the rheology of more and more complex RBC structures (as shown in Fig. 18) with wider

and wider channel widths. In this study, only a single file is observed as there is only one cell in the elementary box. As cell concentration increases, hydrodynamic interactions often cause single-file formations becoming unstable [see Fig. 18(a)], and eventually breaking down into ordered two files [see Fig. 18(b)] or disordered configurations, as previously shown by Thiebaud *et al.* [33] and Shen *et al.* [34]. A remark regarding the 2D and 3D simulations is in order. As there is an extra degree of freedom in 3D, 2D and 3D results could not be compared directly. For example, the swirling motion in 3D is not possible in 2D. However, complex motions such as snaking and chaotic dynamics (see Figs. S3 and S4 in the Supplemental Material [56]) could emerge in different regions of parameter space in 2D and 3D [26,28]. In future works, more attention should be paid in 3D simulations to capture more realistic cell dynamics.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

APPENDIX: INTRINSIC VISCOSITY DUE TO A RIGID CIRCULAR PARTICLE

Consider a rigid circular particle located at the center line of a channel as shown in Fig. 1. The flow is imposed by a pressure gradient

$$\frac{\Delta p_0}{L} = -\frac{8u_{\max}\mu_{\text{ex}}}{W^2}, \tag{A1}$$

and the resulted unperturbed flow velocity is $\boldsymbol{u}_0(u_0, v_0)$, as shown in Eq. (1). The associated stress tensor and dynamic pressure are $\boldsymbol{\sigma}_0$ and p_0. The flow is perturbed by the presence of the particle, and the perturbed flow velocity, stress tensor and pressure are

$$\boldsymbol{u} = \boldsymbol{u}_0 + \boldsymbol{u}', \boldsymbol{\sigma} = \boldsymbol{\sigma}_0 + \boldsymbol{\sigma}', p = p_0 + p'. \tag{A2}$$

At steady state, the particle moves freely and the flow velocity on the particle surface S_p is $\boldsymbol{U}(U_x, 0)$. The boundary conditions on the two wall surfaces S_W are $\boldsymbol{u} = 0$. The remaining control surfaces S_L are the cross sections located at $x = 0$ and $x = L$. Using the Lorentz reciprocity theorem, one can obtain the following equation:

$$\int_{S_p \cup S_L \cup S_W} \boldsymbol{u}_0 \cdot \boldsymbol{\sigma} \cdot \boldsymbol{n} dS = \int_{S_p \cup S_L \cup S_W} \boldsymbol{u} \cdot \boldsymbol{\sigma}_0 \cdot \boldsymbol{n} dS, \tag{A3}$$

where $\boldsymbol{n}$ is the outward normal vector. The integral over the two wall surfaces S_W vanishes due to the no-slip boundary conditions for the unperturbed flow $\boldsymbol{u}_0$ and the perturbed flow $\boldsymbol{u}$. The contributions of the integrals over the particle surface S_p and the cross-sectional control surfaces S_L are derived as follows.

As the particle moves with a constant velocity $\boldsymbol{U}$, the integral of the perturbed flow $\boldsymbol{u}$ and the unperturbed stress tensor $\boldsymbol{\sigma}_0$ over the surface S_p is

$$\int_{S_p} \boldsymbol{u} \cdot \boldsymbol{\sigma}_0 \cdot \boldsymbol{n} dS = \boldsymbol{U} \cdot \int_{S_p} \boldsymbol{\sigma}_0 \cdot \boldsymbol{n} dS = 0, \tag{A4}$$

because the fluid enclosed by the surface S_p is force free in the unperturbed flow. The integral over the control surface S_L is evaluated by decomposing stress tensors $\boldsymbol{\sigma}_0$ and $\boldsymbol{\sigma}$ as their pressure and deviatoric-stress contributions,

$$\boldsymbol{\sigma}_0 = -p_0 \boldsymbol{I} + \boldsymbol{\sigma}_{0d}, \boldsymbol{\sigma} = -p\boldsymbol{I} + \boldsymbol{\sigma}_d. \tag{A5}$$

Due to the use of periodic boundary conditions, the flow velocities on the two cross sections at $x = 0$ and $x = L$ are the same, so do the deviatoric stress tensors. For this reason, the integrals of the deviatoric part vanish as

$$\int_{S_L} \boldsymbol{u} \cdot \boldsymbol{\sigma}_{0d} \cdot \boldsymbol{n} dS = \int_{S_L} \boldsymbol{u}_0 \cdot \boldsymbol{\sigma}_d \cdot \boldsymbol{n} dS = 0. \tag{A6}$$

Recall that the pressure drop remains constant (i.e., $\Delta p_0 = \Delta p$). Then the integrals over the control surface S_L can be expressed as

$$\int_{S_L} \boldsymbol{u} \cdot \boldsymbol{\sigma}_0 \cdot \boldsymbol{n} dS - \int_{S_L} \boldsymbol{u}_0 \cdot \boldsymbol{\sigma} \cdot \boldsymbol{n} dS = Q\Delta p_0 - Q_0 \Delta p = Q' \Delta p_0, \tag{A7}$$

where $Q' = Q - Q_0$ is the excess flux due to the presence of the particle. Substituting Eqs. (A4) and (A7) into the Lorentz reciprocity theorem Eq. (A3), one obtains

$$\int_{S_p} \boldsymbol{u_0} \boldsymbol{F} dS = -Q' \Delta p_0, \tag{A8}$$

where $\boldsymbol{F} = -\boldsymbol{\sigma} \cdot \boldsymbol{n}$ is the surface force on the particle.

The surface force $\boldsymbol{F}$ can be evaluated when hydrodynamic interactions between the particle and walls are neglected. In a polar coordinate system attached to the particle center, the surface force $\boldsymbol{F}(\theta) = -\boldsymbol{\sigma} \cdot \boldsymbol{n}$ is periodic, and can be expressed by Fourier series as

$$F_x(\theta) + iF_y(\theta) = \sum_{k=-\infty}^{\infty} \hat{F}_k e^{ik\theta}, \tag{A9}$$

where $\hat{F}_k$ are the complex amplitudes. Applying force-free and torque-free conditions, one finds

$$\hat{F}_0 = 0, \mathrm{Im}(\hat{F}_1) = 0. \tag{A10}$$

The value of $\mathrm{Re}(\hat{F}_1)$ is set to 0 since it does not affect the flow field. Here Re and Im denote real and imaginary parts, respectively. Neglecting interactions with walls, the particle velocity can be written as

$$\boldsymbol{U} = \boldsymbol{u}_0(\theta) + \frac{1}{4\pi\mu_{\mathrm{ex}}} \int_0^{2\pi} \boldsymbol{G}(\theta, \phi) \boldsymbol{F}(\phi) d\phi, \tag{A11}$$

where $\boldsymbol{G}$ is the free-space Green's function [68]. This equation can be rewritten using Fourier series as

$$U_x = u_{\max} \left(1 - \frac{4R_0^2 \sin^2 \theta}{W^2} \right) + \sum_{k \neq 0,1} \frac{R_0}{4\mu_{\mathrm{ex}}|k|} \hat{F}_k e^{ik\theta}. \tag{A12}$$

Solving the equation for the kth Fourier harmonics, we obtain

$$U_x = u_{\max} \left(1 - \frac{2R_0^2}{W^2} \right), \tag{A13}$$

$$\hat{F}_2 = \hat{F}_{-2} = -\frac{8u_{\max}\mu_{\mathrm{ex}} R_0}{W^2}, \hat{F}_k = 0 \quad \text{for} \quad |k| > 2. \tag{A14}$$

Combining Eqs. (A1), (A8), and (A14), the excess flux Q' is obtained as

$$Q' = -\frac{4\pi u_{\max} R_0^4}{LW^2}. \tag{A15}$$

By substituting Eqs. (8), (10), and (A15) into Eq. (11), the intrinsic viscosity reads

$$[\mu] = \frac{\mu_{\text{eff}} - \mu_{\text{ex}}}{\mu_{\text{ex}}\phi} = -\frac{Q'}{(Q_0 + Q')\phi} = \frac{6R_0^2}{W^2 - 6\pi R_0^4/LW}. \tag{A16}$$

When the channel length L approaches infinity, the intrinsic viscosity tends to $6R_0^2/W^2$.

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